

Wykrywanie zagrożeń i reakcja na incydenty

Laboratorium nr 1

Grupa 5, śr 15:15

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1. Przygotowanie środowiska wirtualnego: najlepiej Virtualbox (crossplatform), ewentualnie ParallelDesktop (jeśli na MacOS)

- Środowisko wirtualne zostało przygotowane zgodnie z wymaganiami. Do jego utworzenia wykorzystano VirtualBox (rozwiązanie crossplatformowe). Podczas konfiguracji zastosowano się do wszystkich zaleceń zawartych w dokumentacji. Ze względu na wcześniejszą pracę z VirtualBoxem, nie posiadano aktualnych zrzutów ekranu przedstawiających sam proces instalacji.

2. Zaopatrzenie się w nośnik ok. 128/256GB (dysk, pendrive) lub miejsce na dysku (oddzielna partycja)

- Zapewniono odpowiednią przestrzeń na dysku w postaci wydzielonej partycji, spełniającą wymagania dotyczące pojemności.

3. Przygotowanie hostów, instalacja, konfiguracja./4. Sprawdzenie domyślnych konfiguracji systemów: otwarte usługi, otwarte porty, kontrola dostępu (firewalle wbudowane)

a) Linux "testujący". Zalecane dystrybucje: Kali Linux, ParrotOS, BlackArch

System Kali Linux został poprawnie zainstalowany i uruchomiony w środowisku wirtualnym. Przeprowadzono weryfikację konfiguracji sieciowej, interfejs działa prawidłowo i przypisano adres IP. Sprawdzono aktywne usługi systemowe oraz konfigurację zapory sieciowej, nie stwierdzono dodatkowych reguł filtrowania. Skanowanie portów nie wykazało otwartych portów, co wskazuje na brak usług nasłuchujących. Zweryfikowano mechanizmy bezpieczeństwa (AppArmor) oraz obecność plików SUID. System znajduje się w stanie domyślnym

Skanowanie portów lokalnych przy użyciu Nmap:

```
(stud@kali)-[~]
└─$ nmap localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 18:41 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds

(stud@kali)-[~]
└─$ nmap -sS -sV localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 18:41 +0100
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000060s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.39 seconds
```

Konfiguracja sieciowa systemu Kali Linux:

```
(stud@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:f6:07:14 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 463sec preferred_lft 463sec
    inet6 fe80::a00:27ff:fef6:714/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Lista aktywnych usług systemowych:

```
(stud@kali)-[~]
$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
accounts-daemon.service             loaded active running Accounts Service
colord.service                       loaded active running Manage, Install and Gene>
cron.service                        loaded active running Regular background progr>
dbus.service                        loaded active running D-Bus System Message Bus
gdm.service                         loaded active running GNOME Display Manager
ModemManager.service               loaded active running Modem Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                      loaded active running Authorization Manager
power-profiles-daemon.service        loaded active running Power Profiles daemon
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-oomd.service                loaded active running Userspace Out-Of-Memory >
systemd-udev.service                loaded active running Rule-based Manager for D>
systemd-userdbd.service             loaded active running User Database Manager
udisks2.service                     loaded active running Disk Manager
upower.service                      loaded active running Daemon for power managem>
user@1000.service                   loaded active running User Manager for UID 1000
virtualbox-guest-utils.service      loaded active running Virtualbox guest utils
wpa_supplicant.service              loaded active running WPA supplicant

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of S>
          SUB     → The low-level unit activation state, values depend on unit typ>

19 loaded units listed.
lines 4-26/26 (END)
cron.service                        loaded active running Regular background program processing daemon
dbus.service                        loaded active running D-Bus System Message Bus
gdm.service                         loaded active running GNOME Display Manager
ModemManager.service               loaded active running Modem Manager
NetworkManager.service             loaded active running Network Manager
polkit.service                      loaded active running Authorization Manager
power-profiles-daemon.service        loaded active running Power Profiles daemon
systemd-journald.service            loaded active running Journal Service
systemd-logind.service              loaded active running User Login Management
systemd-oomd.service                loaded active running Userspace Out-Of-Memory (OOM) Killer
systemd-udev.service                loaded active running Rule-based Manager for Device Events and Files
systemd-userdbd.service             loaded active running User Database Manager
udisks2.service                     loaded active running Disk Manager
upower.service                      loaded active running Daemon for power management
user@1000.service                   loaded active running User Manager for UID 1000
virtualbox-guest-utils.service      loaded active running Virtualbox guest utils
wpa_supplicant.service              loaded active running WPA supplicant
```

Konfiguracja zapory sieciowej:

```
(stud@kali)-[~]
$ sudo iptables -L -n -v
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

(stud@kali)-[~]
```

Lista aktywnych połączeń i portów:

```
(stud@kali)-[~]
$ sudo lsof -i -P -n
COMMAND      PID USER FD   TYPE DEVICE SIZE/OFF NODE NAME
NetworkMa 78294 root 27u  IPv4 130662      0t0  UDP 10.0.2.15:68->10.0.2.2:67

(stud@kali)-[~]
$ arp -a
? (10.0.2.2) at 08:00:27:6f:2f:76 [ether] on eth0
_gateway (10.0.2.1) at 52:55:0a:00:02:01 [ether] on eth0
```

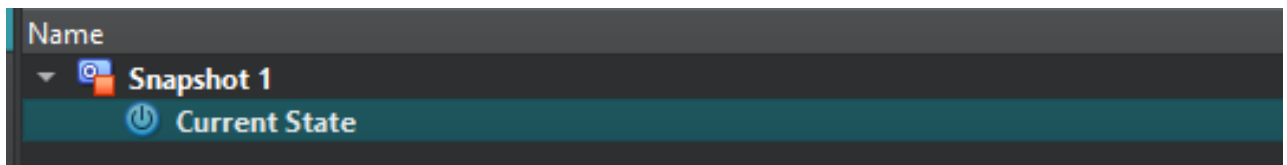
Status mechanizmu bezpieczeństwa AppArmor:

```
(stud@kali)-[~]
$ sudo aa-status
apparmor module is loaded.
116 profiles are loaded.
15 profiles are in enforce mode.
  /usr/bin/man
  /usr/bin/papers
  /usr/bin/papers-previewer
  /usr/bin/papers-previewer//sanitized_helper
  /usr/bin/papers-thumbnailer
  /usr/bin/papers//sanitized_helper
  /usr/bin/papers//snap_browsers
  /usr/libexec/geoclue
```

Lista plików z ustawionym bitem SUID:

```
(stud@kali)-[~]
$ find / -perm -4000 2>/dev/null
/usr/bin/ntfs-3g
/usr/bin/newgrp
/usr/bin/kismet_cap_nrf_52840
```

Po sprawdzeniu i aktualizacji systemu została zrobiona migawka:



b) Linux serwerowy. Zalecane dystrybucje: Ubuntu Server

Została zainstalowana maszyna Ubuntu Server. Wszystkie rzeczy sprawdzone są przedstawione niżej.

Lista aktywnych serwisów:

```
admin@ubuntu-server:~$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
cron.service                       loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
ModemManager.service              loaded active running Modem Manager
multipathd.service                loaded active running Device-Mapper Multipath Device Controller
polkit.service                     loaded active running Authorization Manager
rsyslog.service                   loaded active running System Logging Service
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-networkd.service           loaded active running Network Configuration
systemd-resolved.service           loaded active running Network Name Resolution
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
udisks2.service                   loaded active running Disk Manager
unattended-upgrades.service         loaded active running Unattended Upgrades Shutdown
user@1000.service                  loaded active running User Manager for UID 1000

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

16 loaded units listed.
admin@ubuntu-server:~$
```

Otwarte porty serwera:

```
admin@ubuntu-server:~$ sudo netstat -tulpn
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      458/systemd-resolve
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      458/systemd-resolve
udp        0      0 127.0.0.54:53           0.0.0.0:*               *          458/systemd-resolve
udp        0      0 127.0.0.53:53           0.0.0.0:*               *          458/systemd-resolve
udp        0      0 10.0.2.15:68            0.0.0.0:*               *          833/systemd-network
admin@ubuntu-server:~$ _
```

Dodatkowo skan sieciowy maszyny:

```
admin@ubuntu-server:~$ sudo nmap 10.0.2.15
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-16 17:06 UTC
Nmap scan report for ubuntu-server (10.0.2.15)
Host is up (0.0000020s latency).
All 1000 scanned ports on ubuntu-server (10.0.2.15) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.17 seconds
admin@ubuntu-server:~$ sudo nmap -sS -sV 10.0.2.15
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-03-16 17:06 UTC
Nmap scan report for ubuntu-server (10.0.2.15)
Host is up (0.0000020s latency).
All 1000 scanned ports on ubuntu-server (10.0.2.15) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds
```

Firewall jest wyłączony automatycznie:

```
admin@ubuntu-server:~$ sudo ufw status
Status: inactive
admin@ubuntu-server:~$ _
```

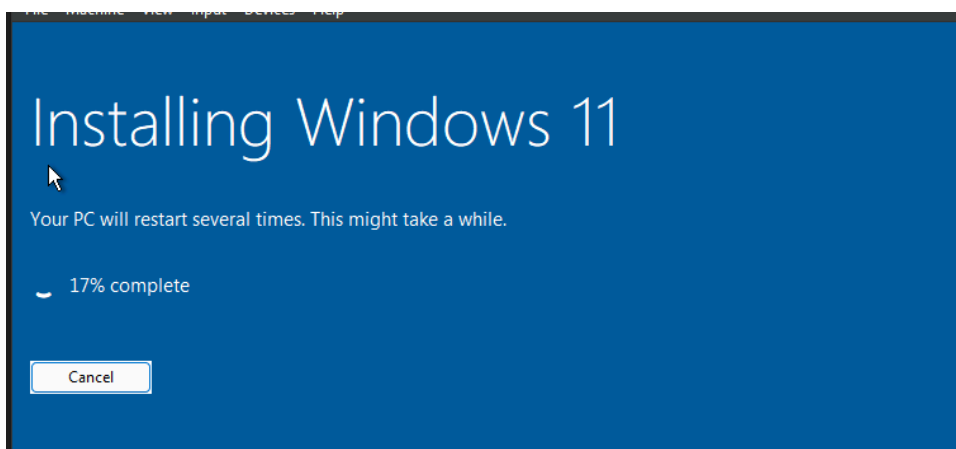
Po sprawdzeniu i aktualizacji systemu została zrobiona migawka:

Name	Taken
Ubuntu_Server_default	3/16/2026 8:27 PM (5 seconds ago)
Current State (changed)	

c) Windows do testowania: Windows 11, standard

System Windows 11 został poprawnie zainstalowany i uruchomiony w środowisku wirtualnym. Przeprowadzono weryfikację konfiguracji sieciowej, system uzyskał adres IP 10.0.2.15, a połączenie sieciowe działa prawidłowo. Sprawdzone aktywne połączenia i porty przy użyciu narzędzia netstat, zidentyfikowano porty nasłuchujące oraz aktywne połączenia wychodzące (m.in. HTTPS). Zweryfikowano konfigurację zapory systemowej, firewall jest włączony dla wszystkich profili i blokuje połączenia przychodzące. Przeanalizowano uruchomione procesy systemowe, które odpowiadają za standardowe działanie systemu operacyjnego. System znajduje się w stanie domyślnym i jest gotowy do dalszych testów.

Proces instalacji systemu Windows 11:



Konfiguracja sieciowa systemu:

```
C:\Users\win>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : t22-8_10.dds.pwr.edu.pl
    IPv6 Address. . . . . : fd17:625c:f037:2:f157:e158:7a0b:f1f
    Temporary IPv6 Address. . . . . : fd17:625c:f037:2:c100:430b:68ae:bb67
    Link-local IPv6 Address . . . . . : fe80::76e8:4a70:5336:f89e%6
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::2%6
                                10.0.2.2
```

Lista aktywnych połączeń i portów:

```
C:\Users\win>netstat -ano

Active Connections

Proto Local Address          Foreign Address         State                   PID
TCP   0.0.0.0:135             0.0.0.0:0               LISTENING               404
TCP   0.0.0.0:445             0.0.0.0:0               LISTENING                4
TCP   0.0.0.0:5040            0.0.0.0:0               LISTENING              5400
TCP   0.0.0.0:7680            0.0.0.0:0               LISTENING             8424
TCP   0.0.0.0:49664           0.0.0.0:0               LISTENING              828
TCP   0.0.0.0:49665           0.0.0.0:0               LISTENING              672
TCP   0.0.0.0:49666           0.0.0.0:0               LISTENING             2040
TCP   0.0.0.0:49668           0.0.0.0:0               LISTENING             2712
TCP   0.0.0.0:49671           0.0.0.0:0               LISTENING             3240
TCP   0.0.0.0:49672           0.0.0.0:0               LISTENING              808
TCP   10.0.2.15:139           0.0.0.0:0               LISTENING                4
TCP   10.0.2.15:49755         172.211.123.250:443     ESTABLISHED            3636
TCP   10.0.2.15:49841         172.211.123.250:443     ESTABLISHED            3636
TCP   10.0.2.15:50242         52.110.19.3:443        ESTABLISHED            7920
TCP   10.0.2.15:50243         52.110.19.1:443        ESTABLISHED            7920
TCP   10.0.2.15:51655         156.17.229.10:80       ESTABLISHED            7408
TCP   10.0.2.15:52401         92.123.189.113:443     CLOSE_WAIT             7408
TCP   10.0.2.15:53689         83.168.92.115:443     ESTABLISHED            5892
TCP   10.0.2.15:53690         83.168.92.107:443     ESTABLISHED            5892
TCP   10.0.2.15:53693         156.17.229.10:80       TIME_WAIT                0
TCP   10.0.2.15:53694         8.8.8.8:443            TIME_WAIT                0
TCP   10.0.2.15:53696         150.171.28.11:443     TIME_WAIT                0
TCP   10.0.2.15:53699         156.17.229.10:80       TIME_WAIT                0
TCP   10.0.2.15:53701         8.8.8.8:443            TIME_WAIT                0
```

Konfiguracja zapory systemowej:

```
C:\Users\win>netsh advfirewall show allprofiles

Domain Profile Settings:
-----
State                               ON
Firewall Policy                     BlockInbound,AllowOutbound
LocalFirewallRules                  N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096

Private Profile Settings:
-----
State                               ON
Firewall Policy                     BlockInbound,AllowOutbound
LocalFirewallRules                  N/A (GPO-store only)
LocalConSecRules                     N/A (GPO-store only)
InboundUserNotification              Enable
RemoteManagement                    Disable
UnicastResponseToMulticast           Enable

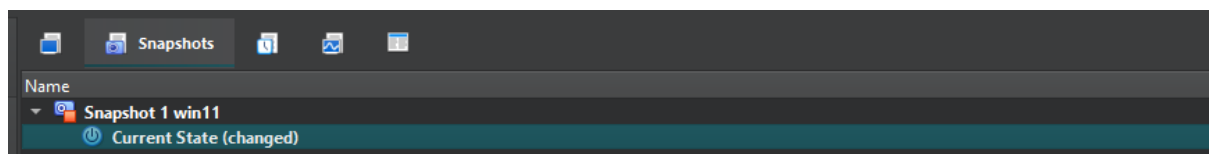
Logging:
LogAllowedConnections                Disable
LogDroppedConnections                Disable
FileName                             %systemroot%\system32\LogFiles\Firewall\pfirewall.log
MaxFileSize                          4096
```

Lista uruchomionych procesów :

```
C:\Users\win>tasklist
```

Image Name	PID	Session Name	Session#	Mem Usage
System Idle Process	0	Services	0	8 K
System	4	Services	0	168 K
Registry	96	Services	0	22,196 K
smss.exe	416	Services	0	1,676 K
csrss.exe	600	Services	0	7,436 K
wininit.exe	672	Services	0	9,520 K
csrss.exe	680	Console	1	7,540 K
winlogon.exe	740	Console	1	18,864 K
services.exe	808	Services	0	11,556 K
lsass.exe	828	Services	0	31,888 K
svchost.exe	952	Services	0	37,208 K
fontdrvhost.exe	980	Services	0	4,764 K
fontdrvhost.exe	988	Console	1	6,264 K
svchost.exe	404	Services	0	18,576 K
svchost.exe	388	Services	0	10,312 K
dwm.exe	528	Console	1	117,636 K
svchost.exe	1060	Services	0	5,524 K
svchost.exe	1128	Services	0	8,104 K
svchost.exe	1212	Services	0	9,224 K
svchost.exe	1220	Services	0	16,444 K
svchost.exe	1300	Services	0	12,892 K
svchost.exe	1528	Services	0	11,476 K
svchost.exe	1568	Services	0	18,484 K
svchost.exe	1636	Services	0	29,608 K
svchost.exe	1712	Services	0	10,660 K
svchost.exe	1772	Services	0	20,396 K
svchost.exe	2012	Services	0	11,204 K
svchost.exe	2040	Services	0	16,800 K
svchost.exe	440	Services	0	7,204 K
svchost.exe	1044	Services	0	24,756 K
svchost.exe	2076	Services	0	13,244 K

Wykonano również snapshot maszyny wirtualnej w celu umożliwienia przywrócenia początkowej konfiguracji systemu.



d) Host standalone: Tails

Wybrana została Tails OS, postawiona w Virtualboxie, żeby zachować zmiany nie była wyłączona tylko został zachowany jej stan. Po aktualizacji systemu zostały zweryfikowana konfiguracja standardowa systemu.

Wyświetlone aktywne serwisy:

```
amnesia@amnesia:~$ systemctl list-units --type=service --state=running
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
colord.service	loaded	active	running	Manage, Install
cron.service	loaded	active	running	Regular backgro
cups.service	loaded	active	running	CUPS Scheduler
dbus.service	loaded	active	running	D-Bus System Me
gdm.service	loaded	active	running	GNOME Display M
haveged.service	loaded	active	running	Entropy Daemon
memlockd.service	loaded	active	running	memlockd
ModemManager.service	loaded	active	running	Modem Manager
NetworkManager-dispatcher.service	loaded	active	running	Network Manager
NetworkManager.service	loaded	active	running	Network Manager
onion-grater.service	loaded	active	running	Tor control por
polkit.service	loaded	active	running	Authorization M
rtkit-daemon.service	loaded	active	running	RealtimeKit Sch
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Mana
systemd-oomd.service	loaded	active	running	Userspace Out-0
systemd-udevd.service	loaded	active	running	Rule-based Mana
tails-detect-disk-ioerrors.service	loaded	active	running	Detect Squashfs
tails-persistent-storage.service	loaded	active	running	tails-persisten
tails-shutdown-on-media-removal.service	loaded	active	running	Wipe memory on
tca-portal.service	loaded	active	running	Provides privil

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
cron.service	loaded	active	running	Regular background program processing daemon
cups.service	loaded	active	running	CUPS Scheduler
dbus.service	loaded	active	running	D-Bus System Message Bus
gdm.service	loaded	active	running	GNOME Display Manager
haveged.service	loaded	active	running	Entropy Daemon based on the HAVEGE algorithm
memlockd.service	loaded	active	running	memlockd
ModemManager.service	loaded	active	running	Modem Manager
NetworkManager-dispatcher.service	loaded	active	running	Network Manager Script Dispatcher Service
NetworkManager.service	loaded	active	running	Network Manager
onion-grater.service	loaded	active	running	Tor control port filter proxy
polkit.service	loaded	active	running	Authorization Manager
rtkit-daemon.service	loaded	active	running	RealtimeKit Scheduling Policy Service
systemd-journald.service	loaded	active	running	Journal Service
systemd-logind.service	loaded	active	running	User Login Management
systemd-oomd.service	loaded	active	running	Userspace Out-Of-Memory (OOM) Killer
systemd-udevd.service	loaded	active	running	Rule-based Manager for Device Events and Files
tails-detect-disk-ioerrors.service	loaded	active	running	Detect Squashfs and I/O errors
tails-persistent-storage.service	loaded	active	running	tails-persistent-storage.service
tails-shutdown-on-media-removal.service	loaded	active	running	Wipe memory on live media removal
tca-portal.service	loaded	active	running	Provides privileged functions for TCA
tor@default.service	loaded	active	running	Anonymizing overlay network for TCP
udisks2.service	loaded	active	running	Disk Manager
upower.service	loaded	active	running	Daemon for power management
usbguard-dbus.service	loaded	active	running	USBGuard D-Bus Service
usbguard.service	loaded	active	running	USBGuard daemon
user@1000.service	loaded	active	running	User Manager for UID 1000
wpa_supplicant.service	loaded	active	running	WPA supplicant

Legend: LOAD → Reflects whether the unit definition was properly loaded.
ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
SUB → The low-level unit activation state, values depend on unit type.

29 loaded units listed.

Otwarte porty:

```
root@amnesia:/home/amnesia# ss -tulpn
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 127.0.0.1:5353 0.0.0.0:* users:(("tor",pid=11835,fd=10))
tcp LISTEN 0 4096 127.0.0.1:9052 0.0.0.0:* users:(("tor",pid=11835,fd=12))
tcp LISTEN 0 4096 127.0.0.1:9050 0.0.0.0:* users:(("tor",pid=11835,fd=6))
tcp LISTEN 0 4096 127.0.0.1:9040 0.0.0.0:* users:(("tor",pid=11835,fd=11))
tcp LISTEN 0 4096 127.0.0.1:39787 0.0.0.0:* users:(("obfs4proxy",pid=11855,fd=6))
tcp LISTEN 0 4096 127.0.0.1:9062 0.0.0.0:* users:(("tor",pid=11835,fd=7))
tcp LISTEN 0 4096 127.0.0.1:9063 0.0.0.0:* users:(("tor",pid=11835,fd=8))
tcp LISTEN 0 5 0.0.0.0:951 0.0.0.0:* users:(("onion-grater",pid=4882,fd=3))
tcp LISTEN 0 4096 127.0.0.1:631 0.0.0.0:* users:(("cupsd",pid=5190,fd=6))
tcp LISTEN 0 4096 10.200.1.1:9050 0.0.0.0:* users:(("tor",pid=11835,fd=9))
tcp LISTEN 0 4096 127.0.0.1:38423 0.0.0.0:* users:(("obfs4proxy",pid=11855,fd=8))
tcp LISTEN 0 4096 127.0.0.1:34255 0.0.0.0:* users:(("obfs4proxy",pid=11855,fd=9))
tcp LISTEN 0 4096 127.0.0.1:41999 0.0.0.0:* users:(("obfs4proxy",pid=11855,fd=7))
root@amnesia:/home/amnesia#
```

Uruchomiony firewall z regułami:

```
root@amnesia:/home/amnesia# iptables -L
Chain INPUT (policy DROP)
target prot opt source destination state
ACCEPT all -- anywhere anywhere state ESTABLISHED
ACCEPT all -- anywhere anywhere
ACCEPT tcp -- 10.200.1.2 10.200.1.1 multiport dports 9050,951
ACCEPT tcp -- 10.200.1.6 10.200.1.5 multiport dports 951
ACCEPT tcp -- 10.200.1.10 10.200.1.9 multiport dports 951
ACCEPT tcp -- 10.200.1.14 10.200.1.13 multiport dports 9050,951
REJECT all -- anywhere 10.200.1.0/24 reject-with icmp-port-unreachable

Chain FORWARD (policy DROP)
target prot opt source destination
ACCEPT all -- anywhere anywhere
ACCEPT all -- anywhere anywhere

Chain OUTPUT (policy DROP)
target prot opt source destination
ACCEPT all -- anywhere anywhere state ESTABLISHED
ACCEPT icmp -- anywhere anywhere state RELATED
ACCEPT tcp -- anywhere localhost tcp dpt:9050 flags:FIN,SYN,RST,ACK/SYN owner UID match _apt
ACCEPT tcp -- anywhere localhost tcp dpt:9050 flags:FIN,SYN,RST,ACK/SYN owner UID match proxy
ACCEPT tcp -- anywhere localhost tcp dpt:9050 flags:FIN,SYN,RST,ACK/SYN owner UID match nobody
ACCEPT tcp -- anywhere localhost tcp dpt:9050 flags:FIN,SYN,RST,ACK/SYN owner UID match tails-upgrade-frontend
ACCEPT tcp -- anywhere localhost tcp dpt:9062 flags:FIN,SYN,RST,ACK/SYN owner UID match http
ACCEPT tcp -- anywhere localhost tcp dpt:9062 flags:FIN,SYN,RST,ACK/SYN owner UID match tails-upgrade-frontend
ACCEPT tcp -- anywhere localhost tcp dpt:9063 flags:FIN,SYN,RST,ACK/SYN owner UID match tails-iuk-get-target-fi
log_reject tcp -- anywhere localhost tcp dpt:9063 flags:FIN,SYN,RST,ACK/SYN owner UID match amnesia
log_reject tcp -- anywhere localhost tcp dpt:9052 owner UID match amnesia
ACCEPT tcp -- anywhere localhost tcp dpt:9052 owner UID match root
ACCEPT tcp -- anywhere localhost tcp dpt:951 owner UID match root
ACCEPT udp -- anywhere localhost udp dpt:domain owner UID match amnesia
ACCEPT udp -- anywhere localhost udp dpt:mdns owner UID match amnesia
ACCEPT udp -- anywhere localhost udp dpt:domain owner UID match http
ACCEPT udp -- anywhere localhost udp dpt:mdns owner UID match http
log_reject udp -- anywhere localhost udp dpt:domain owner UID match _apt
log_reject udp -- anywhere localhost udp dpt:mdns owner UID match _apt

log_reject tcp -- anywhere localhost tcp dpt:9063 flags:FIN,SYN,RST,ACK/SYN owner UID match amnesia
log_reject tcp -- anywhere localhost tcp dpt:9052 owner UID match amnesia
ACCEPT tcp -- anywhere localhost tcp dpt:9052 owner UID match root
ACCEPT tcp -- anywhere localhost tcp dpt:951 owner UID match root
ACCEPT udp -- anywhere localhost udp dpt:domain owner UID match amnesia
ACCEPT udp -- anywhere localhost udp dpt:mdns owner UID match amnesia
ACCEPT udp -- anywhere localhost udp dpt:domain owner UID match http
ACCEPT udp -- anywhere localhost udp dpt:mdns owner UID match http
log_reject udp -- anywhere localhost udp dpt:domain owner UID match _apt
log_reject udp -- anywhere localhost udp dpt:mdns owner UID match _apt
ACCEPT udp -- anywhere localhost udp dpt:domain owner UID match root
ACCEPT udp -- anywhere localhost udp dpt:mdns owner UID match root
ACCEPT tcp -- anywhere localhost tcp dpt:4101 flags:FIN,SYN,RST,ACK/SYN owner UID match Debian-gdm
ACCEPT tcp -- anywhere localhost tcp flags:FIN,SYN,RST,ACK/SYN owner UID match amnesia
ACCEPT tcp -- anywhere anywhere owner UID match clearnet
ACCEPT tcp -- anywhere anywhere owner UID match debian-tor tcp flags:FIN,SYN,RST,ACK/SYN state NEW
ACCEPT udp -- anywhere anywhere owner UID match debian-tor udp dpt:domain
lan all -- anywhere 10.0.0.0/8
lan all -- anywhere 172.16.0.0/12
lan all -- anywhere 192.168.0.0/16
log_reject all -- anywhere anywhere

Chain lan (3 references)
target prot opt source destination
log_reject tcp -- anywhere anywhere tcp dpt:domain
log_reject udp -- anywhere anywhere udp dpt:domain
log_reject udp -- anywhere anywhere udp dpt:netbios-ns
ACCEPT all -- anywhere anywhere

Chain log_reject (8 references)
target prot opt source destination
LOG all -- anywhere anywhere LOG level debug uid prefix "Dropped outbound packet: "
REJECT all -- anywhere anywhere reject-with icmp-port-unreachable
root@amnesia:/home/amnesia#
```

Skan wszystkich interfejsów Tails OS:

```
root@amnesia:/home/amnesia# nmap -sS -sV 10.200.1.0/30
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-16 19:17 UTC
sendto in send_ip_packet_sd: sendto(6, packet, 44, 0, 10.200.1.2, 16) => Operation not permitted
Offending packet: TCP 10.200.1.1:49260 > 10.200.1.2:53 S ttl=55 id=4290 iplen=44 seq=1233587430 win=1024 <mss 1460>
sendto in send_ip_packet_sd: sendto(6, packet, 44, 0, 10.200.1.2, 16) => Operation not permitted
Offending packet: TCP 10.200.1.1:49262 > 10.200.1.2:53 S ttl=37 id=18963 iplen=44 seq=1233456356 win=1024 <mss 1460>
Nmap scan report for 10.200.1.2
Host is up (0.0000050s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
53/tcp    filtered  domain
MAC Address: 86:DE:B8:EA:5D:70 (Unknown)

sendto in send_ip_packet_sd: sendto(6, packet, 44, 0, 10.200.1.1, 16) => Operation not permitted
Offending packet: TCP 10.200.1.1:49516 > 10.200.1.1:53 S ttl=59 id=48929 iplen=44 seq=3783734343 win=1024 <mss 1460>
Nmap scan report for 10.200.1.1
Host is up (0.000019s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE      SERVICE VERSION
53/tcp    filtered  domain
9050/tcp  open      tor-socks Tor SOCKS proxy

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 4 IP addresses (2 hosts up) scanned in 8.88 seconds
root@amnesia:/home/amnesia#
```

Po sprawdzeniu i aktualizacji systemu została zrobiona migawka:

Name	Taken
▼  Tails_default	3/16/2026 8:27 PM (27 seconds ago)
 Current State	